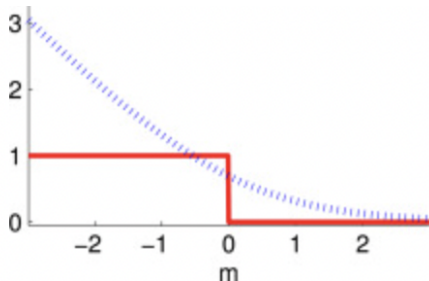


Regularisation

Procheta Sen

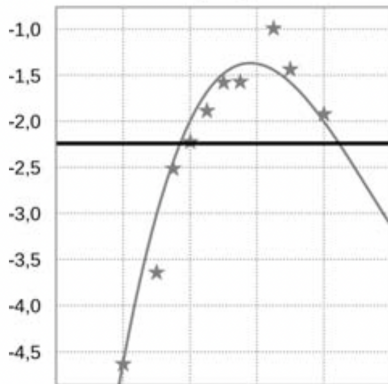


Introduction to Regularisation

- ▶ Regularisation is a process of reducing overfitting in a model by constraining it (reducing the complexity/no. of parameters).
- ▶ For classifiers that use a weight vector, regularisation can be done by minimising the norm (length) of the weight vector.
- ▶ Several popular regularisation methods exist.
 - ▶ L2 regularisation (ridge regression or Tikhonov regularisation).
 - ▶ L1 regularisation (Lasso regression).
 - ▶ L1+L2 regularisation (mixed regularisation).

Approximation of a Polynomial d

- ▶ Approximation of $f(x) = x^3 - 4x^2 + 3x - 2$ by a polynomial of degree d .
- ▶ The dataset $\mathcal{D} = \{x_i, y_i\}_{i=1}^n$ is sampled from the polynomial $f(x) = x^3 - 4x^2 + 3x - 2$ with some noise.



Approximation Loss Function

- ▶ We want to approximate the unknown function $f(x)$ by a polynomial of degree d .
- ▶ $\hat{y}(x, \overline{W}) = W_0 + \sum_{j=1}^d w_j x^j = (1, x^1, x^2 \dots x^d) \cdot \overline{W}$
- ▶ such that the following loss function (residual sum of squares (RSS)) is minimised
$$L(\mathcal{D}, \overline{W}) = \sum_{i=1}^n (\hat{y}(x_i, \overline{W}) - y_i)^2$$

Approximation Example

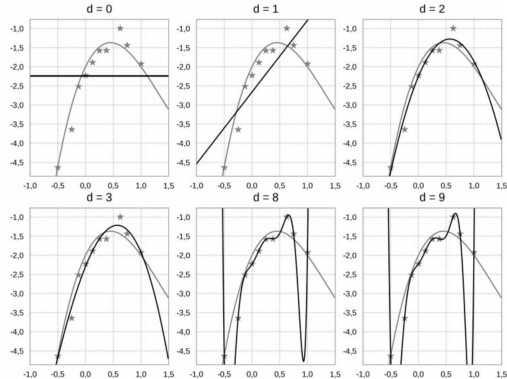


Figure: Approximation with different values of d

Approximation Formulations

- ▶ $f_0(x) = -2,2393$
- ▶ $f_1(x) = -2,6617 + 1,8775x$
- ▶ $f_2(x) = -2,2528 + 3,4604x - 3,0603x^2$
- ▶ $f_8(x) = -2,2324 + 2,2326x + 6,2543x^2 + 15,599gx^3 - 239,9751x^4 + 322,8516x^4 + 621,0952x^6 - 1478,6505x^7 + 750,9032x^8$
- ▶ $f_9(x) = -2,22 + 2,01x + 4,88x^2 + 31,13x^3 - 230,31x^4 + 103,72x^5 + 869,22x^6 - 966,67x^7 - 319,31x^8, 505,64x^9$
- ▶ The parameters grow!
- ▶ Let's try to restrict their growth.
- ▶ $L(\mathcal{D}, \overline{W}) = \sum_{i=1}^n (\hat{y}(x_i, \overline{W}) - y_i)^2 + \lambda ||W||^2$

Approximation with Different values of λ

$$f_{\lambda=0}(x) = -2,22 + 2,01x + 4,88x^2 + 31,13x^3 - 230,31x^4 + \\ + 103,72x^5 + 869,22x^6 - 966,67x^7 - 319,31x^8 + 505,64x^9,$$

$$f_{\lambda=0,01}(x) = -2,32 + 3,40x - 2,33x^2 + 0,05x^3 - 0,51x^4 - \\ - 0,29x^5 - 0,22x^6 - 0,06x^7 + 0,09x^8 + 0,24x^9,$$

$$f_{\lambda=1}(x) = -2,46 + 1,45x - 0,19x^2 + 0,22x^3 - 0,13x^4 - \\ - 0,05x^5 - 0,14x^6 - 0,13x^7 - 0,16x^8 - 0,16x^9.$$

Figure: Equation with different values of λ

$f(x)$ Plot With Different Values of Lambda

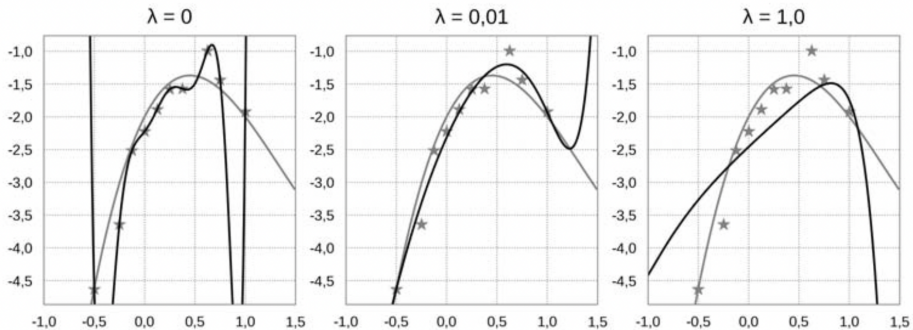


Figure: Caption

L2 Regularisation

- ▶ Let us denote by $L(\mathcal{D}, \mathbf{W})$ the loss of classifying the dataset $\mathcal{D}$ using the model represented by the weight vector $\mathbf{W}$.
- ▶ We would like to impose L2 regularisation on $\mathbf{W}$.
- ▶ The overall objective to minimise can then be written as follows:

$$J(\mathcal{D}, \mathbf{W}) = L(\mathcal{D}, \mathbf{W}) + \lambda \|\mathbf{W}\|^2 = L(\mathcal{D}, \mathbf{W}) + \lambda \sum_{i=1}^d w_i^2$$

- ▶ Here λ is called the **regularisation coefficient** and is usually set via **cross-validation**.
- ▶ The gradient of the overall objective simply becomes the sum of the loss-gradient and the scaled weight vector $\mathbf{W}$.

Examples

- ▶ Note that SGD update rule for minimising a loss multiplies the loss gradient by a negative learning rate (μ).
- ▶ Therefore, the L2 regularised update rules will have a $-2\mu\lambda\mathbf{W}$ term as shown in the following examples.

L2 Regularised Perceptron Update Rule

$$\mathbf{W} \leftarrow \mathbf{W} - \mu (-y_i \mathbf{X}_i + 2\lambda \mathbf{W})$$

$$= \mathbf{W} + \mu y_i \mathbf{X}_i - 2\mu \lambda \mathbf{W}$$

$$= \mathbf{W} + y_i \mathbf{X}_i - 2\lambda \mathbf{W} \quad (\text{for } \mu = 1)$$

$$= (1 - 2\lambda) \mathbf{W} + y_i \mathbf{X}_i$$

How to set λ

- ▶ Split your training dataset into **training** and **validation** parts (e.g., 80%–20%).
- ▶ Try different values for λ (typically in the logarithmic scale), e.g.

$$\lambda = 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 10^2, 10^3, 10^4, 10^5$$

- ▶ Train a different classification model for each λ and select the value that gives the best performance (e.g., accuracy) on the validation data.